Functional specialisation in the first years of life: longitudinal characterisation of social perception with fNIRS

J Benerradi [a], C Bulgarelli [b], A Blasi [c], B Blanco [a], S McCann [d], B Milosavljevic [e], E Mbye [f], E Touray [f], S Moore [d, f], C Elwell [c], S Lloyd-Fox [a], The BRIGHT Team

[a] Psychology, University of Cambridge, UK; [b] Psychological Sciences, Birkbeck University of London, UK; [c] Medical Physics & Biomedical Engineering, University College London, UK; [d] Women & Children's Health, King's College London, UK; [e] Biological and Behavioural Sciences, Queen Mary University of London, UK; [f] The Medical Research Council Unit The Gambia at the London School of Hygiene and Tropical Medicine, The Gambia

Presenting Author email address: jb2615@cam.ac.uk

Abstract: By studying brain responses to social perception longitudinally in infants and children in The Gambia, we found that social vocal stimuli elicited a significantly slower and stronger response than auditory non-vocal stimuli across all 6 age points collected in the Brain Imaging for Global Health (BRIGHT) project from 5 months to 5 years old. We show the advantage of studying brain activation informed by the speed and magnitude of the response with functional near-infrared spectroscopy (fNIRS) to characterise the brain response more comprehensively.

Introduction: Infants and children in low- and middle-income countries have historically been under-represented in neurodevelopmental research, despite being often exposed to early adversity which can impact cognitive development. The Brain Imaging for Global Health (BRIGHT) project enabled the collection of a large dataset for studying cognitive development longitudinally from birth to 5 years old in The Gambia and UK[1]. While social processing has been studied often in infants younger than 1 year of age in high-income countries, we here present findings with functional near-infrared spectroscopy (fNIRS) on a social task in the Gambian cohort from 5 to 60 months.

Methods: Participants were presented with social videos accompanied by vocal and non-vocal sounds following a block design; a study paradigm common in previous literature and adapted to the local context[2]. Participants were tested at 5 months (*N*=127), 8 months (*N*=110), 12 months (*N*=116), 18 months (*N*=111), 24 months (*N*=112) and 60 months (*N*=124) with fNIRS from the temporal and inferior frontal brain regions. Channels with low scalp coupling were pruned and motion artefacts corrected using spline interpolation and wavelet filtering[3]. The data was converted into oxy- (HbO) and deoxy-haemoglobin (HbR), and blocks were averaged for each condition (vocal or non-vocal). We studied longitudinally the average time-to-peak (speed) and peak amplitude (magnitude) of the haemodynamic response function (HRF), for each condition, which informed the appropriate time window to selected for statistical analysis on HbO and HbR to investigate brain activation.

Results: We found a significantly greater activation in response to vocal compared to non-vocal stimuli across all ages, located bilaterally in temporal regions but also in the inferior frontal gyri. This becomes more widespread with age, with activation on 8 out of 34 channels at 5 months compared to 21 out of 34 at 60 months on this vocal minus non-vocal contrast. Furthermore, HbO and HbR reached their peak amplitude significantly slower in response to social vocal compared to non-vocal stimuli (*Fig. 1*) consistently from 5 to 60 months (all p-values < 0.05 except HbR at 60 months where p-value < 0.1), with the absolute amplitude of the peak also appearing to decrease with age up to 24 months.

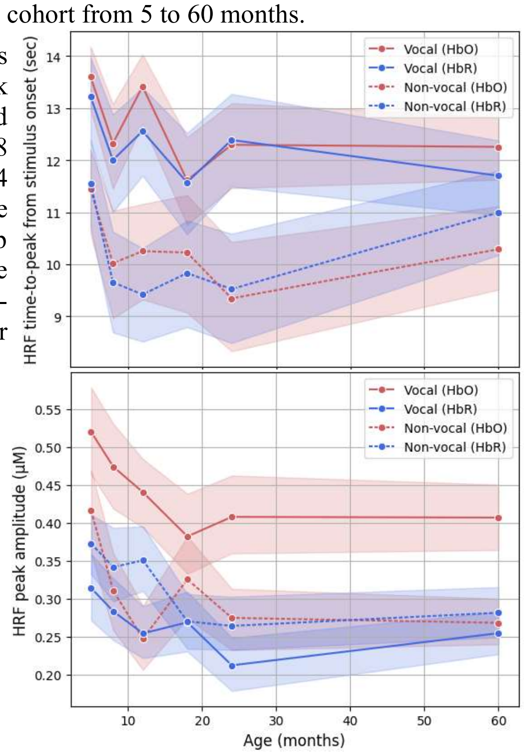

Fig. 1: Characteristics of the HRF across ages

Conclusion: Studying the speed and magnitude of the response with fNIRS enabled to have more details about the brain responses to social perception in infants and children longitudinally, but also to inform the time window on which to focus statistical analysis. Ongoing work is now investigating whether amplitude and time-to-peak change as a function of the cortical location to gain further insight into the dynamics of the response across ages.

1. S Lloyd-Fox et al., 2023 - 2. S Lloyd-Fox et al., 2014 - 3. C Bulgarelli et al., 2020